1.18

Using invariants: $\vec{E}\cdot\vec{B} = \vec{E}'\cdot\vec{B}' = E'B'$

$\uparrow$ frame K $\quad \uparrow$ K'

$$E' = \frac{\vec{E}\cdot\vec{B}}{B'}. \quad \text{Then} \quad F_{\mu\nu}F^{\mu\nu} \sim B^2 - \frac{E^2}{c^2} = B'^2 - \frac{E'^2}{c^2}$$

$$\Rightarrow c^2B^2 - E^2 = c^2B'^2 - \frac{(\vec{E}\cdot\vec{B})^2}{B'^2}$$

$$c^2B'^4 - (c^2B^2 - E^2)B'^2 - (\vec{E}\cdot\vec{B})^2 = 0$$

$$\Rightarrow B'^2 = \frac{1}{2c^2}\left(c^2B^2 - E^2 \pm \sqrt{(c^2B^2-E^2)^2 + 4c^2(\vec{E}\cdot\vec{B})^2}\right)$$

B' is real $\Rightarrow$ sign is $\oplus$.

$$= \frac{1}{2c^2}(2c\,\vec{E}\cdot\vec{B})\left\{\sqrt{\left(\frac{E^2-c^2B^2}{2c\,\vec{E}\cdot\vec{B}}\right)^2+1} - \frac{E^2-c^2B^2}{2c\,\vec{E}\cdot\vec{B}}\right\}$$

Denote $C = \dfrac{E^2-c^2B^2}{2c\,\vec{E}\cdot\vec{B}}$

$$\Rightarrow cB' = \sqrt{c\,\vec{E}\cdot\vec{B}}\left[\sqrt{1+C^2} - C\right]^{1/2} = \left[\frac{c\,\vec{E}\cdot\vec{B}}{\sqrt{1+C^2}+C}\right]^{1/2}.$$

Consider $\dfrac{1}{\sqrt{1+C^2}+C} = \dfrac{\sqrt{1+C^2}-C}{1+C^2-C^2} = \sqrt{1+C^2}-C$

Then $E' = \dfrac{c\,\vec{E}\cdot\vec{B}}{cB'} = \left[c\,\vec{E}\cdot\vec{B}\left(\sqrt{1+C^2}+C\right)\right]^{1/2}.$

Interesting related problem: find velocity $\mathbf{v}$ of K'.

In K: $$\left(\vec{E}\times\vec{B}\right)^2 = \left(EB\sin\theta\right)^2 = E^2B^2\left(1-\cos^2\theta\right) = E^2B^2 - \left(\vec{E}\cdot\vec{B}\right)^2$$

In K': $\left(\vec{E}'\times\vec{B}'\right)^2 = 0$ ← this defines frame K'. K' moves along axis $\hat{x} = \dfrac{\vec{E}\times\vec{B}}{|\vec{E}\times\vec{B}|}$.

But $$\left(\vec{E}'\times\vec{B}'\right)^2 = E'^2B'^2 - \underbrace{\left(\vec{E}'\cdot\vec{B}'\right)^2}_{\left(\vec{E}\cdot\vec{B}\right)^2} = 0$$ since $\vec{E}\cdot\vec{B} = \vec{E}'\cdot\vec{B}'$.

$$E'_y = \gamma\left(E_y - vB_z\right) \qquad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right)$$
$$E'_z = \gamma\left(E_z + vB_y\right) \qquad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

$$\Rightarrow \left(\vec{E}'\right)^2 = E'^2 = \gamma^2\left[\left(E_y - vB_z\right)^2 + \left(E_z + vB_y\right)^2\right]$$

$$\left(\vec{B}'\right)^2 = B'^2 = \gamma^2\left[\left(B_y + \frac{v}{c^2}E_z\right)^2 + \left(B_z - \frac{v}{c^2}E_y\right)^2\right]$$
$$= \gamma^2\left[B_y^2 + B_z^2 + \frac{\beta^2}{c^2}\left(E_z^2 + E_y^2\right) + \frac{2\beta}{c}\left(B_yE_z - B_zE_y\right)\right]$$

and $$\left(\vec{E}'\right)^2 = \gamma^2\left[E_y^2 + E_z^2 + v^2\left(B_y^2 + B_z^2\right) + 2v\left(B_yE_z - B_zE_y\right)\right]$$
$$= \gamma^2\left[E_y^2 + E_z^2 + c^2\beta^2\left(B_y^2 + B_z^2\right) - 2c\beta\,\vec{E}\times\vec{B}\cdot\hat{x}\right]$$
$$= \gamma^2\left[E^2 + \beta^2c^2B^2 - 2c\beta\,\vec{E}\times\vec{B}\cdot\hat{x}\right]$$

$$\left(\vec{B}'\right)^2 = \frac{\gamma^2}{c^2}\left[\beta^2E^2 + c^2B^2 - 2c\beta\,\vec{E}\times\vec{B}\cdot\hat{x}\right]$$

Now, $$E'^2B'^2 = \left(\vec{E}'\cdot\vec{B}'\right)^2$$

$$\Rightarrow \frac{\gamma^4}{c^2}\cancel{\left[E^2 + \beta^2c^2B^2 - 2\beta c\,\vec{E}\times\vec{B}\cdot\hat{x}\right]\left[\beta^2E^2 + c^2B^2 - 2\beta c\,\vec{E}\times\vec{B}\cdot\hat{x}\right]} =$$
$$\cancel{= \left(\vec{E}\cdot\vec{B}\right)^2}$$

Thus $(E')^2(B')^2 = (\bar{E}\cdot\bar{B})^2 = (E)^2(B^2) - (\vec{E}\times\vec{B})^2$.

$$\frac{\gamma^4}{c^2}\left(E^2+\beta^2c^2B^2-2c\beta\,\bar{E}\times\bar{B}\cdot\hat{x}\right)\left(\beta^2E^2+c^2B^2-2c\beta\,\bar{E}\times\bar{B}\cdot\hat{x}\right)=$$

$$= E^2B^2-(\bar{E}\times\bar{B})^2$$

$$\left(E^2+\beta^2c^2B^2\right)\left(\beta^2E^2+c^2B^2\right) + 4c^2\beta^2(\bar{E}\times\bar{B})^2$$

$$-2c\beta\,\bar{E}\times\bar{B}\cdot\hat{x}\left(\beta^2E^2+c^2B^2+E^2+\beta^2c^2B^2\right)=$$

$$= c^2(1-\beta^2)^2\left(E^2B^2-(\bar{E}\times\bar{B})^2\right)$$

$$(\beta^4+1)E^2c^2B^2 + \beta^2(E^4+c^4B^4) + (\bar{E}\times\bar{B})^2c^2\left(4\beta^2+(1-\beta^2)^2\right)$$

$$-2c\beta\,\bar{E}\times\bar{B}\cdot\hat{x}(1+\beta^2)(E^2+c^2B^2) = c^2(1-\beta^2)^2E^2B^2$$

But $\beta^4+1-(1-\beta^2)^2 = \beta^4+1-(1-2\beta^2+\beta^4) = 2\beta^2$.

$4\beta^2+(1-\beta^2)^2 = (1+\beta^2)^2$

$\Rightarrow$ everything simplifies as

$$2\beta^2E^2c^2B^2 + \beta^2(E^4+c^4B^4) + (1+\beta^2)^2c^2(\bar{E}\times\bar{B})^2 =$$

$$= 2c\beta(1+\beta^2)(E^2+c^2B^2)\,\bar{E}\times\bar{B}\cdot\hat{x}.$$

$$\Rightarrow \quad \beta^2(E^2+c^2B^2)^2 - 2c\beta(1+\beta^2)(E^2+c^2B^2)\,\bar{E}\times\bar{B}\cdot\hat{x} +$$

$$+ (1+\beta^2)^2c^2(\bar{E}\times\bar{B})^2 = 0 \quad \text{full square!}$$

$$\Rightarrow \left[\beta(E^2+c^2B^2) - (1+\beta^2)\,c\,\bar{E}\times\bar{B}\cdot\hat{x}\right]^2 = 0$$

$$\frac{\beta}{1+\beta^2} = \frac{c\,\bar{E}\times\bar{B}\cdot\hat{x}}{E^2+c^2B^2} \quad \text{or} \quad \frac{1}{\beta}+\beta = \frac{E^2+c^2B^2}{c(\vec{E}\times\vec{B}\cdot\hat{x})}$$

Denote $A = \dfrac{E^2 + c^2 B^2}{c\,\bar{E}\times\bar{B}\cdot x} \Rightarrow \beta + \dfrac{1}{\beta} = A$

$$\beta^2 - A\beta + 1 = 0 \Rightarrow \beta = \frac{A \pm \sqrt{A^2 - 4}}{2}$$

How to choose sign? For $\bar{E}\times\bar{B} = 0$ we should get $\beta = 0$. But in this limit $A \to \infty \Rightarrow$ the sign must be $(-)$.

$$\beta = \frac{1}{2}\left\{\frac{E^2 + c^2 B}{c\,\bar{E}\times\bar{B}\cdot\hat{x}} - \sqrt{\left(\frac{E^2 + c^2 B^2}{\bar{E}\times\bar{B}c}\right)^2 - 4}\right\} =$$

$$= \frac{1}{2c\,\bar{E}\times\bar{B}\cdot\hat{x}}\left\{E^2 + c^2 B^2 - \sqrt{(E^2 + c^2 B^2)^2 - 4(\bar{E}\times\bar{B})^2 c^2}\right\}$$

$$= \frac{\vec{E}\times\vec{B}\cdot\hat{x}}{2c(\vec{E}\times\vec{B})^2}\left\{E^2 + c^2 B^2 - \sqrt{(E^2 - c^2 B^2)^2 + 4c^2(\vec{E}\cdot\vec{B})^2}\right\}$$

using $(\vec{E}\times\vec{B})^2 = \vec{E}^2\vec{B}^2 - (\vec{E}\cdot\vec{B})^2$.

Note that $2c\beta\,\vec{E}\times\vec{B}\cdot\hat{x} = E^2 + c^2 B^2 - \sqrt{(E^2 - c^2 B^2)^2 + 4c^2(\bar{E}\cdot\bar{B})^2}$

$$\Rightarrow (\vec{E}')^2 = \gamma^2\left(E^2 + \beta^2 c^2 B^2 - 2c\beta\,\bar{E}\times\bar{B}\cdot\hat{x}\right) =$$

$$= \gamma^2\left\{(\beta^2 - 1)c^2 B^2 + \sqrt{(E^2 - c^2 B^2)^2 + 4c^2(\bar{E}\cdot\bar{B})^2}\right\}$$

$$c^2(\vec{B}')^2 = \gamma^2\left\{(\beta^2 - 1)E^2 + \sqrt{(E^2 - c^2 B^2)^2 + 4c^2(\bar{E}\cdot\bar{B})^2}\right\}.$$

1.19

K': $\vec{J}' = \sigma \vec{E}'$, $\vec{E}' \approx \underbrace{\vec{E} + \vec{v} \times \vec{B}}_{\text{frame } K}$

Consider $J^\alpha = \frac{1}{c^2}(U_\beta J^\beta) U^\alpha + \sigma F^{\alpha\beta} U_\beta$

$U_\beta = \gamma(c, -\vec{v})$, $J^\beta = (\rho c, \vec{J})$

$U_\beta J^\beta = \gamma(\rho c^2 - \vec{v} \cdot \vec{J}) = \gamma(\rho c^2 - \sigma \vec{v} \cdot \vec{E}) \approx \rho c^2$ in non-relativistic limit.

$\Rightarrow \frac{1}{c^2}(U_\beta J^\beta) U^\alpha \approx \rho \gamma (c, \vec{v}) \approx (c\rho, \rho \vec{v})$.

$$F^{\alpha\beta} U_\beta = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix} \begin{pmatrix} \gamma c \\ -\gamma v_x \\ -\gamma v_y \\ -\gamma v_z \end{pmatrix}$$

$\Rightarrow F^{0\beta} U_\beta = \frac{\gamma}{c} \vec{E} \cdot \vec{v}$;

$F^{1\beta} U_\beta = \gamma(E_x + v_y B_z - v_z B_y) = \gamma(\vec{E} + \vec{v} \times \vec{B})_x$

$\Rightarrow$ we obtain

$$J^\alpha = (c\rho, \vec{v}\rho) + \sigma\gamma\left(\frac{\vec{E} \cdot \vec{v}}{c}, \vec{E} + \vec{v} \times \vec{B}\right)$$

$\begin{pmatrix} \gamma \to 1 \\ \frac{v}{c} \ll 1 \end{pmatrix}$

$$\approx (c\rho, \vec{v}\rho) + \sigma(0, \vec{E} + \vec{v} \times \vec{B})$$

$$= (c\rho, \vec{v}\rho + \sigma \vec{E} + \vec{v} \times \vec{B})$$ as (1.214) says.

1.20

K': $A'^{\alpha} = \left(\frac{\varphi'}{c}, \vec{A}'\right) \qquad \varphi' = \frac{q}{4\pi\varepsilon_0 r'}, \quad \vec{A}' = 0$

K:

$$\begin{pmatrix} \frac{\varphi}{c} \\ A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \varphi'/c \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma \frac{\varphi'}{c} \\ \gamma\beta \frac{\varphi'}{c} \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \varphi(\vec{r}) = \gamma \varphi'(\vec{r}')$$

$$A_x(\vec{r}) = \frac{v}{c^2} \gamma \varphi' = \frac{v}{c^2} \varphi(\vec{r}')$$

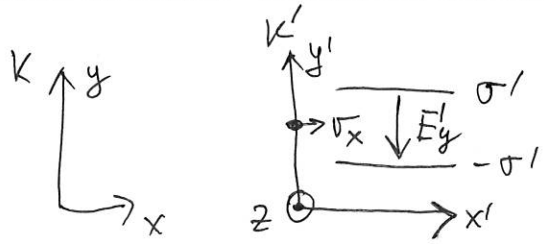
But the argument $\vec{r}'$ transforms as well

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \gamma(ct - \beta x) \\ \gamma(x - \beta ct) \\ y \\ z \end{pmatrix}$$

$$\Rightarrow r' = \sqrt{(x')^2 + (y')^2 + (z')^2} = \sqrt{\gamma^2 (x - vt)^2 + y^2 + z^2}$$

$$\Rightarrow \varphi(\vec{r}) = \frac{\gamma q}{4\pi\varepsilon_0 \left[\gamma^2 (x - vt)^2 + y^2 + z^2\right]^{1/2}}$$

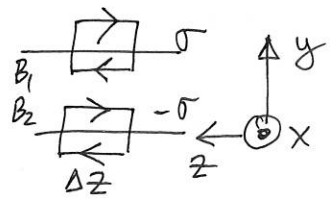
$$A_x(\vec{r}) = \frac{v}{c^2} \varphi(\vec{r}), \quad A_y = A_z = 0.$$



$$E'_y = \frac{\sigma'}{2\varepsilon_0}\cdot 2 = \frac{\sigma'}{\varepsilon_0}$$

$$\vec{E}' = \left(0, -\frac{\sigma}{\varepsilon_0}, 0\right)$$

In the lab frame (frame K) capacitor moves towards us (out of plane)



$$dx'dy'dz' = \gamma\, dx\, dy\, dz$$

$$\Rightarrow \sigma = \frac{q}{A} = \frac{q}{\frac{A'}{\gamma}} = \frac{\gamma q}{A'} = \gamma\sigma'$$

Surface charge density increases!

Magnetic field between capacitor's plates is due to the current $v_x \sigma \Delta z$ by the top plate and $v_x(-\sigma)\Delta z$ by the bottom one.
In between the plates the two fields just add up (and outside they cancel out)

$\boxed{\text{get sign by right-hand rule}}$ $\int \vec{B}_1 \cdot d\vec{l} = \mu_0 I_1 \Rightarrow -2B_1 \Delta z = \mu_0 v_x \sigma \Delta z$

$$B_1 = -\frac{1}{2}\mu_0 v_x \sigma$$

$$\int \vec{B}_2 \cdot d\vec{l} = \mu_0 I_2 \Rightarrow +2B_2 \Delta z = \mu_0 v_x(-\sigma)\Delta z$$

$$B_2 = -\frac{1}{2}\mu_0 v_x \sigma$$

$$\Rightarrow B_{tot} = B_z = -\mu_0 v_x \sigma = -\mu_0 v_x \gamma \sigma' = -\mu_0 v_x \gamma \varepsilon_0 E'_y$$

$$= +\gamma \frac{v_x}{c^2} E'_y$$

Lorentz transformation gives

$$B_z = \gamma\left(B'_z + \frac{v}{c^2}E'_y\right) = \gamma \frac{v}{c^2} E'_y \quad \text{since } B'_z = 0.$$

The two expressions agree